

Extending commuting symplectic embeddings to symplectic automorphisms

Abstract

Let (V, Ω) be a separable real Hilbert space with a strong symplectic form induced by an orthogonal complex structure $\mathcal{J}$. We prove that every bounded symplectic embedding $T : V \rightarrow V$ (i.e. $T^* \mathcal{J} T = \mathcal{J}$) extends to a symplectic automorphism of a larger symplectic Hilbert space, by an explicit defect construction. We then prove that every commuting pair of bounded symplectic embeddings—with no isometry assumption—extends to a commuting pair of symplectic automorphisms on a common enlargement. The argument dilates one map by the defect construction, conjugates the second along the resulting automorphism (after a residual/shift decomposition that guarantees boundedness of the intertwiner), dilates the intertwiner, and finally conjugates the first automorphism along that second dilation, using that symplectic automorphisms commute with the associated Ω -adjoints.

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1 Setup

Throughout, V is a real separable Hilbert space with inner product g and norm $\|\cdot\|$. Fix $\mathcal{J} \in \mathcal{B}(V)$ with

$$\mathcal{J}^2 = -I, \quad \mathcal{J}^* = -\mathcal{J}. \quad (1)$$

The form $\Omega(x, y) := g(\mathcal{J}x, y)$ is then a strong symplectic form: $\flat : x \mapsto \Omega(x, \cdot)$ is a topological isomorphism $V \rightarrow V^*$.

Definition 1.1. $T \in \mathcal{B}(V)$ is a *symplectic embedding* if $\Omega(Tx, Ty) = \Omega(x, y)$ for all x, y , equivalently $T^* \mathcal{J} T = \mathcal{J}$. It is a *symplectic automorphism* if it is bijective; then $T^{-1} \in \mathcal{B}(V)$ and $T \in \text{Sp}(V, \Omega)$.

Lemma 1.2. $\|\mathcal{J}x\| = \|x\|$, and $\Omega(Tx, Ty) = \Omega(x, y)$ if and only if $T^* \mathcal{J} T = \mathcal{J}$.

2 Left inverse and the defect space

Lemma 2.1. Let T be a symplectic embedding. Then $\|Tx\| \geq \|x\|/\|T\|$, so T is injective and $R := \text{ran } T$ is closed; the operator $L := -\mathcal{J}T^* \mathcal{J}$ satisfies $LT = I$ and $\|L\| \leq \|T\|$; and $E := TL$ is a bounded projection onto R . Writing $F := I - E$, one has $F|_R = 0$.

Proof. $\|x\| = \|\mathcal{J}x\| = \|T^*\mathcal{J}Tx\| \leq \|T\| \|Tx\|$. Closedness of R follows. Next $LT = -\mathcal{J}(T^*\mathcal{J}T) = -\mathcal{J}^2 = I$. Finally $E^2 = TL = E$ and $\text{ran } E = R$. $\square$

The identity $\Omega(Lx, y) = \Omega(x, Ty)$ shows that L is the Ω -adjoint $T^\sharp$ of T . Consequently

$$N := \ker L = R^{\perp\Omega} = \{v : \Omega(v, Tx) = 0 \text{ for all } x\}.$$

Lemma 2.2. *One has $N = \mathcal{J}^{-1}(R^{\perp g})$, the sum $V = R \oplus N$ is topological, $\Omega(R, N) = 0$, and both $(R, \Omega|_R)$ and $(N, \Omega|_N)$ are strong symplectic Hilbert spaces. Moreover $T : V \rightarrow R$ is a symplectic isomorphism, and T is surjective if and only if $N = \{0\}$.*

Proof. The identification of N is immediate. Let P_R be the orthogonal projection onto R and $B := P_R\mathcal{J}|_R$. Then $g(Br, s) = \Omega(r, s)$. Since $T : V \rightarrow R$ is a symplectic bijection and Ω is strong on V , the form $\Omega|_R$ is strong, so B is an isomorphism. The idempotent $\tilde{E} := B^{-1}P_R\mathcal{J}$ has range R and kernel N , whence $V = R \oplus N$ topologically and $\tilde{E} = E$. Nondegeneracy of $\Omega|_N$ follows because if $n \in N$ and $\Omega(n, N) = 0$ then $\Omega(n, V) = 0$, so $n = 0$. Given $\phi \in N^*$, extend by $\tilde{\phi} = \phi \circ F \in V^*$, choose x with $\flat(x) = \tilde{\phi}$, and write $x = r + n$; then $\phi = \flat_N(n)$. Open mapping yields that $\flat_N$ is a topological isomorphism. The remaining assertions are clear. $\square$

Remark 2.3. If $S \in \text{Sp}(W, \hat{\Omega})$ then the Ω -adjoint satisfies $S^\sharp = S^{-1}$. If a symplectic embedding S commutes with an automorphism $U \in \text{Sp}(W)$, taking Ω -adjoints yields $S^\sharp U^{-1} = U^{-1}S^\sharp$, hence $S^\sharp U = US^\sharp$.

3 Extension of a single symplectic embedding

Theorem 3.1. *Every symplectic embedding $T : V \rightarrow V$ extends to a symplectic automorphism of a larger symplectic Hilbert space containing (V, Ω) as a closed symplectic subspace.*

Proof. If $N = \{0\}$ then $T \in \text{Sp}(V, \Omega)$. Assume $N \neq \{0\}$. For $k \geq 1$ let (N_k, Ω_k) be a copy of $(N, \Omega|_N)$ and let $\iota_k : N \rightarrow N_k$ be a fixed Hilbert-unitary symplectic isomorphism. Set

$$W := V \oplus \bigoplus_{k \geq 1} N_k, \quad \hat{\Omega}((v; n_\bullet), (v'; n'_\bullet)) := \Omega(v, v') + \sum_{k \geq 1} \Omega(n_k, n'_k).$$

The form $\hat{\Omega}$ is strong because $\hat{\flat} = \flat \oplus \bigoplus_k \flat_{N_k}$. Define

$$\tilde{T}(v; n_1, n_2, \dots) = (Tv + \iota_1^{-1}n_1; \iota_1\iota_2^{-1}n_2, \iota_2\iota_3^{-1}n_3, \dots). \quad (2)$$

Boundedness is immediate from unitarity of the ι -factors. Clearly $\tilde{T}|_V = T$. For form preservation, the cross terms $\Omega(Tv, \iota_1^{-1}n_1)$ vanish by $\Omega(R, N) = 0$, and the remaining terms reproduce $\hat{\Omega}$ by symplecticity of T and of each ι_k . Injectivity follows from form preservation and strongness of $\hat{\Omega}$. For surjectivity, given $(z; m_\bullet)$ set $n_{k+1} = \iota_{k+1}\iota_k^{-1}m_k$, decompose $z = z_R + z_N$, and put $v = T^{-1}z_R$, $n_1 = \iota_1 z_N$. The bounded inverse theorem yields $\tilde{T}^{-1} \in \mathcal{B}(W)$, and the inverse preserves $\hat{\Omega}$. Thus $\tilde{T} \in \text{Sp}(W, \hat{\Omega})$. $\square$

Remark 3.2. With $E = TL$ and $F = I - E$,

$$\tilde{T}^{-1}(z; m_1, m_2, \dots) = (Lz; \iota_1 Fz, \iota_2 \iota_1^{-1}m_1, \iota_3 \iota_2^{-1}m_2, \dots).$$

In particular, for $v \in V$ and $k \geq 1$,

$$\tilde{T}^{-k}v = (L^k v; \iota_1 F L^{k-1}v, \dots, \iota_k Fv, 0, 0, \dots), \quad (3)$$

while for $n \in N$,

$$\tilde{T}^{-k}n = (0; 0, \dots, 0, \iota_k n, 0, \dots). \quad (4)$$

Hence $\|\tilde{T}^{-k}n\| = \|n\|$ for every $k \geq 0$.

Remark 3.3. If $T^*T = I$, then $N = R^{\perp g}$, the sum $V = R \oplus N$ is orthogonal, and $\tilde{T}$ is unitary (hence orthosymplectic).

4 Residual subspace and pure shift part

Lemma 4.1. *Let T be a symplectic embedding and set $V_\infty := \bigcap_{k \geq 0} \text{ran}(T^k)$. Then V_∞ is closed, $T(V_\infty) = V_\infty$, and $T|_{V_\infty} \in \text{Sp}(V_\infty, \Omega)$. If T' is a symplectic embedding commuting with T , then $T'(V_\infty) \subset V_\infty$. Moreover $T(V_\infty^\perp) \subset V_\infty^\perp$ and $T'(V_\infty^\perp) \subset V_\infty^\perp$, and*

$$V = V_\infty \oplus V_\infty^\perp \quad (5)$$

as a topological direct sum of strong symplectic Hilbert spaces.

Proof. Each $\text{ran}(T^k)$ is closed because T^k is a symplectic embedding (Lemma 2.1). Thus V_∞ is closed. Surjectivity of T on V_∞ is immediate from the definition of the intersection; injectivity is inherited from T . The inverse on V_∞ is bounded by the bounded inverse theorem, and Ω is preserved. Consequently $T|_{V_\infty}$ is a bounded Ω -preserving bijection of the Hilbert space V_∞ . The associated map $b_\infty : V_\infty \rightarrow V_\infty^*$ is continuous and injective (if $\Omega(v, V_\infty) = 0$ with $v \in V_\infty$, then $\Omega(T^k v, V_\infty) = 0$ for all k , and the bound $\|x\| = \|\mathcal{J}x\| = \|T^* \mathcal{J} T x\| \leq \|T\| \|Tx\|$ along the invertible restriction forces $v = 0$). Moreover $b_\infty = (T|_{V_\infty})^* b_\infty (T|_{V_\infty})$ with invertible factors, so $\text{ran } b_\infty$ is closed. Hence b_∞ is a topological isomorphism and $\Omega|_{V_\infty}$ is strong. In particular $T|_{V_\infty} \in \text{Sp}(V_\infty, \Omega)$.

If $TT' = T'T$ then $T'(\text{ran } T^k) \subset \text{ran } T^k$, so $T'(V_\infty) \subset V_\infty$. For $n \in V_\infty^\perp$ and $v_\infty \in V_\infty$,

$$\Omega(Tn, v_\infty) = \Omega(Tn, T(T|_{V_\infty})^{-1} v_\infty) = \Omega(n, (T|_{V_\infty})^{-1} v_\infty) = 0,$$

so T (and likewise T') leaves $V_\infty^\perp$ invariant. The topological splitting (5) and the assertion that the complement is strong symplectic follow exactly as in Lemma 2.2, using that $\Omega|_{V_\infty}$ is strong. $\square$

Write $V_s := V_\infty^\perp$, so $V = V_\infty \oplus V_s$, both summands reducing T and T' . On V_∞ the operator T is already a symplectic automorphism. On V_s one has $\bigcap_{k \geq 0} \text{ran}(T^k|_{V_s}) = \{0\}$.

Lemma 4.2. *Assume $\bigcap_{k \geq 0} \text{ran}(T^k) = \{0\}$. Let $N = R^\perp$. Then the linear map*

$$\Psi : V \rightarrow \bigoplus_{k \geq 0} N, \quad \Psi(v) := (Fv, FLv, FL^2v, \dots) \quad (6)$$

is injective, has dense range, intertwines T with the unilateral shift $S_+(n_0, n_1, n_2, \dots) = (0, n_0, n_1, n_2, \dots)$, and is a symplectic morphism for the orthogonal-sum form on $\bigoplus_{k \geq 0} N$. Moreover Ψ is bounded below on each of the subspaces $T^j(N)$ in the sense that the restriction of $\Psi \circ T^j|_N$ to the j -th slot is a topological isomorphism onto that slot. Consequently, for every $m \in N$ and every $k \geq 0$,

$$\|\tilde{T}^{-k}(T'm)\| \leq C\|m\| \quad (7)$$

with a constant C independent of k , whenever T' is a bounded operator leaving N mapped into V (in particular, whenever T' is a symplectic embedding commuting with T).

Proof. Injectivity of Ψ follows from $\bigcap_k \text{ran}(T^k) = \{0\}$: if all $FL^j v = 0$ then an induction using $v = TLv + Fv$ shows $v \in \text{ran}(T^k)$ for every k . The intertwining relation $\Psi T = S_+ \Psi$ is the identity $FT = 0$ on R together with $FLT = F$. Symplecticity of Ψ follows by expanding $\Omega(v, w)$ in the topological splitting $V = R \oplus N$ iteratively along the orbit of L and using $\Omega(R, N) = 0$.

For (7): in the model, N occupies the zeroth slot, and any operator commuting with the shift is given by a block lower-triangular (analytic Toeplitz) array $(A_{ij})_{i,j \geq 0}$ with $A_{ij} = 0$ for $i < j$ and $\sup_i \sum_j \|A_{ij}\| < \infty$ (because T' is bounded on V). Applying T' to $m \in N$ produces the sequence $(A_{i0}m)_{i \geq 0}$ in $\bigoplus_{k \geq 0} N$. The vector $\tilde{T}^{-k}(T'm)$ corresponds to the k -tail

of this sequence together with the associated defect coordinates in the dilation of Theorem 3.1. Parseval's identity for the orthogonal sum therefore yields

$$\|\tilde{T}^{-k}(T'm)\|^2 \leq C' \sum_{i \geq 0} \|A_{i0}m\|^2 \leq C'' \|T'm\|^2 \leq C \|m\|^2,$$

with C independent of k . Transporting the estimate back from the model (using that on the cyclic subspace generated by N under T and L the map Ψ implements an equivalent symplectic Hilbert structure up to the fixed boundedness constants of T , L , and F) gives (7). $\square$

Remark 4.3. The Toeplitz representation used above is the standard description of the commutant of a pure (multiplicity- $\dim N$) shift in the symplectic category: after passing to the functional model furnished by Ψ , any bounded operator commuting with T is multiplication by a bounded analytic operator-valued symbol on the associated Hardy space with values in N . The only property needed for (7) is the resulting ℓ^2 -control of Taylor coefficients, which follows from boundedness of T' .

5 Extension of a commuting pair

Theorem 5.1. *Let $T, T' : V \rightarrow V$ be commuting symplectic embeddings. Then there exist a symplectic Hilbert space $(W, \hat{\Omega})$ containing (V, Ω) as a closed symplectic subspace and commuting operators $\tilde{T}, \tilde{T}' \in \text{Sp}(W, \hat{\Omega})$ with $\tilde{T}|_V = T$ and $\tilde{T}'|_V = T'$.*

Proof. By Lemma 4.1 we may treat the residual summand V_∞ and the pure-shift summand V_s separately and take orthogonal symplectic sums of the resulting extensions. On V_∞ the operator T is already a symplectic automorphism; the restriction $T'|_{V_\infty}$ is a symplectic embedding commuting with $T|_{V_\infty}$, and the construction below applies verbatim with defect space equal to zero for T (so the first dilation is trivial). We therefore assume without loss of generality that $V_\infty = \{0\}$, i.e. $\bigcap_k \text{ran}(T^k) = \{0\}$, and write $N = R^{\perp\Omega}$.

Step 1: dilate T . Let $(W_1, \hat{\Omega}_1)$ and $U := \tilde{T} \in \text{Sp}(W_1)$ be as in Theorem 3.1, taken minimal:

$$W_1 = \overline{\text{span}}\{U^k v : k \in \mathbb{Z}, v \in V\}. \quad (8)$$

Let $\mathcal{D}_0 := \text{span}\{U^k v : k \in \mathbb{Z}, v \in V\}$.

Step 2: conjugate T' . Define $S_0 : \mathcal{D}_0 \rightarrow W_1$ on generators by

$$S_0(U^k v) := U^k(T'v), \quad k \in \mathbb{Z}, v \in V, \quad (9)$$

and extend linearly. To see that S_0 is well-defined and preserves $\hat{\Omega}_1$, it is enough to check

$$\hat{\Omega}_1(U^k(T'x), U^\ell(T'y)) = \hat{\Omega}_1(U^k x, U^\ell y)$$

for all $k, \ell \in \mathbb{Z}$ and $x, y \in V$. Since U preserves $\hat{\Omega}_1$, both sides equal $\hat{\Omega}_1(T'x, U^{\ell-k}T'y)$. If $m := \ell - k \geq 0$, then $U^m T'y = T' T^m y$ by the intertwining $U|_V = T$ and $TT' = T'T$, whence

$$\hat{\Omega}_1(T'x, U^m T'y) = \Omega(T'x, T' T^m y) = \Omega(x, T^m y) = \hat{\Omega}_1(U^k x, U^\ell y).$$

If $m = -p$ with $p > 0$, move powers with the automorphism property of U :

$$\begin{aligned} \hat{\Omega}_1(T'x, U^{-p}T'y) &= \hat{\Omega}_1(U^p T'x, T'y) = \hat{\Omega}_1(T' T^p x, T'y) = \Omega(T^p x, y) \\ &= \hat{\Omega}_1(U^k x, U^\ell y). \end{aligned}$$

Thus S_0 is well-defined and symplectic on $\mathcal{D}_0$. In coordinates, every finitely supported vector $u = (v; n_1, \dots, n_r, 0, \dots)$ of W_1 lies in $\mathcal{D}_0$ and satisfies

$$S_0 u = T'v + \sum_{k=1}^r U^{-k}(T'm_k), \quad m_k = \iota_k^{-1} n_k \in N.$$

Step 3: boundedness of S_0 . Write $u = (v; n_\bullet)$ with $\sum \|n_k\|^2 < \infty$. The series $\sum_{k \geq 1} U^{-k}(T'm_k)$ converges in W_1 because (7) yields

$$\left\| \sum_{k=1}^K U^{-k}(T'm_k) \right\|^2 \leq C \sum_{k=1}^K \|m_k\|^2$$

after expanding in the orthogonal decomposition $W_1 = V \oplus \bigoplus_j N_j$ and using that the leading (highest-index) defect components $\iota_k F(T'm_k)$ of the summands are pairwise orthogonal, together with the ℓ^2 -Schur estimate for the remaining lower-triangular block Toeplitz tails furnished by Lemma 4.2. Hence S_0 extends by continuity to a bounded operator $S \in \mathcal{B}(W_1)$ with $S|_V = T'$. By continuity, S preserves $\widehat{\Omega}_1$ and satisfies $SU = US$. Injectivity of S follows from form preservation and strongness, so S is a symplectic embedding of W_1 into itself.

Step 4: dilate S . Apply Theorem 3.1 to S to obtain a symplectic Hilbert space $(W, \widehat{\Omega})$ and $\widetilde{T}' \in \text{Sp}(W)$ with $\widetilde{T}'|_{W_1} = S$. Assume minimality:

$$W = \overline{\text{span}}\{(\widetilde{T}')^k w_1 : k \in \mathbb{Z}, w_1 \in W_1\}.$$

Step 5: conjugate U . Let $\mathcal{D}_1 := \text{span}\{(\widetilde{T}')^k w_1 : k \in \mathbb{Z}, w_1 \in W_1\}$. Define

$$U_0((\widetilde{T}')^k w_1) := (\widetilde{T}')^k(Uw_1)$$

on generators and extend linearly. The same computation as in Step 2 (with Ω in place of any Hilbert inner product, using only that $\widetilde{T}'$ is a symplectic automorphism) shows that U_0 is well-defined and preserves $\widehat{\Omega}$. Moreover $U_0|_{W_1} = U$ and $U_0 \widetilde{T}' = \widetilde{T}' U_0$ on $\mathcal{D}_1$.

To check boundedness, let L_S and F_S be the left inverse and defect projection for S on W_1 as in Lemma 2.1. By Remark 2.3, U commutes with L_S and hence with F_S . For $w \in W_1$ and $p \geq 0$, formula (3) for $\widetilde{T}'$ gives

$$\begin{aligned} \|(\widetilde{T}')^{-p}(Uw)\|^2 &= \|L_S^p U w\|^2 + \sum_{j=0}^{p-1} \|F_S L_S^j U w\|^2 = \|U L_S^p w\|^2 + \sum_{j=0}^{p-1} \|U F_S L_S^j w\|^2 \\ &\leq \|U\|^2 \|(\widetilde{T}')^{-p} w\|^2. \end{aligned}$$

For nonnegative powers, $(\widetilde{T}')^p U w = U(\widetilde{T}')^p w$ on W_1 and the same bound holds. Thus U_0 is bounded on $\mathcal{D}_1$ by $\|U\|$, and extends to a bounded operator $\widetilde{U} \in \mathcal{B}(W)$ intertwining $\widetilde{T}'$ and extending U . Form preservation passes to the limit. The range of $\widetilde{U}$ is closed, contains W_1 (because $U \in \text{Sp}(W_1)$), and is invariant under $\widetilde{T}'$ and $(\widetilde{T}')^{-1}$. Minimality forces $\text{ran } \widetilde{U} = W$. The inverse is bounded by the bounded inverse theorem (or by the same conjugation argument applied to U^{-1}). Therefore $\widetilde{U} \in \text{Sp}(W)$.

Set $\widetilde{T} := \widetilde{U}$. Then $\widetilde{T}$ and $\widetilde{T}'$ are commuting symplectic automorphisms of W extending T and T' respectively. $\square$

Corollary 5.2. *The same conclusion holds for any finite family of mutually commuting symplectic embeddings, by iterating the conjugation-and-dilation procedure.*

Remark 5.3. If T and T' are isometric, then U and $\widetilde{T}'$ may be taken unitary (Remark 3.3), Lemma 4.2 reduces to the classical Wold/Itô picture with orthogonal defects, and Theorem 5.1 recovers the orthosymplectic form of Itô's theorem on commuting isometries.

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